DynoSLAM: Dynamic SLAM with Generative Graph Neural Networks for Real-World Social Navigation

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Research Questions

- How can Graph Neural Networks be integrated directly into GraphSLAM factor graph optimization to handle dynamic pedestrians?
- How does stochastic motion prediction with Monte Carlo rollouts compare to deterministic approaches in avoiding the 'argmax problem' in social navigation?
- Can empirical covariance from GNN rollouts provide mathematically rigorous uncertainty quantification for downstream planning algorithms?

Core Idea

Traditional SLAM assumes static environments, but fails catastrophically with moving pedestrians due to corrupted optimization from dynamic landmarks. Deterministic neural predictors suffer from the 'argmax problem' - when facing ambiguous social situations (e.g., pass left vs right), they average outcomes into unphysical straight-line trajectories. DynoSLAM solves this by treating a Graph Attention Network as a stochastic World Model, using Monte Carlo rollouts with small perturbations to sample multimodal human intent distributions. The key insight is embedding this uncertainty directly into SLAM via dynamic Mahalanobis distance factors that automatically adjust stiffness - tight binding when predictions are confident, relaxed when interactions are ambiguous.

Key Formula

Stochastic GAT Mahalanobis Loss

$$L_{kin}^{(k,i)} = \|m_{k,i} - m_{k,i-1} - \mu_{k,i}\Delta t\|^2_{\Sigma_{k,i}^{-1}}$$

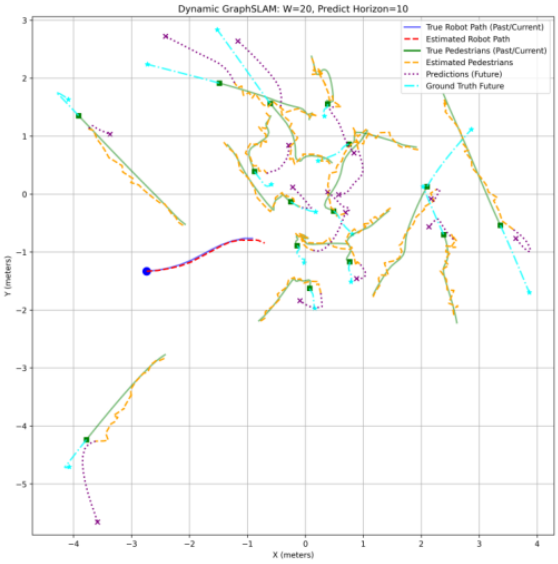
m_{k,i} is landmark position, \mu_{k,i} is empirical mean velocity from Monte Carlo samples, \Sigma_{k,i} is empirical covariance matrix. This factor dynamically modulates kinematic prior stiffness - small covariance creates strong binding when predictions are confident, large covariance relaxes constraints during ambiguous interactions.

Empirical Covariance from Monte Carlo Rollouts

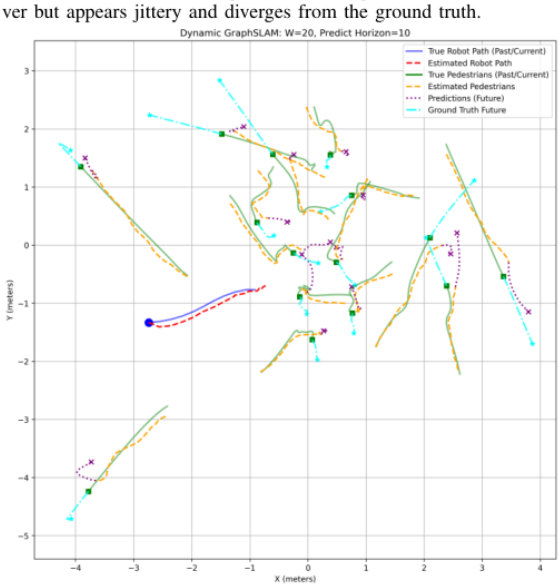
$$\Sigma_{k,i} = \frac{1}{N-1}\sum_{s=1}^{N}(\tilde{v}_{k,i}^{(s)} - \mu_{k,i})(\tilde{v}_{k,i}^{(s)} - \mu_{k,i})^T$$

\tilde{v}_{k,i}^{(s)} are velocity samples from perturbed GAT rollouts, \mu_{k,i} is sample mean, N is number of Monte Carlo samples. This captures epistemic uncertainty of social interactions - concentrated samples yield small covariance in unambiguous scenarios, spread samples yield large covariance during complex interactions.

Key Figure



(a) Deterministic GAT: The prediction attempts an avoidance maneuver but appears jittery and diverges from the ground truth.



(b) Stochastic GAT (Ours): By averaging multiple Monte Carlo rollouts and applying the Mahalanobis loss, the prediction becomes smooth and aligns perfectly with the true future.

Fig. 4: Qualitative comparison of multi-agent prediction horizons within the dynamic SLAM graph.

provides downstream local planners (such as MPC or PPO) with a strict probabilistic safety envelope. Rather than blindly trusting a single maximum-likelihood trajectory, the control algorithm can optimize the robot's actions to safely avoid the entire distribution of potential non-linear human maneuvers.

V. Conclusion

In this work, we introduced a novel Dynamic GraphSLAM framework tightly coupled with socially-aware, generative motion priors. Our experiments on simulated highly-interactive environments demonstrate that traditional static SLAM approaches and Constant Velocity baselines fail to anticipate the complex, non-linear avoidance maneuvers characteristic of human crowds.

We showed that while single-agent neural networks (MLP) collapse to linear extrapolation due to a lack of spatial context, integrating a Graph Attention Network (GAT) effectively captures pedestrian social dynamics. Most crucially, we proved that framing the motion predictor as a stochastic World Model—generating multiple future hypotheses rather than a single deterministic output—fundamentally improves the system. By mapping the variance of these rollouts into a Mahalanobis distance within the SLAM factor graph, the system achieves dynamic stiffness, yielding highly accurate trajectory tracking (Robot RMSE of 0.39m) while explicitly quantifying interaction uncertainty.

Furthermore, this dynamic tracking system is designed to complement existing static SLAM pipelines. By running in parallel, our framework can refine the poses of moving and static landmarks and the robot simultaneously, effectively cleaning the dynamic noise that usually corrupts static mapping. Ultimately, the continuous extraction of multi-agent states and uncertainty covariance matrices provides an ideal input space for safety-critical control algorithms like Model Predictive Control (MPC), enabling anticipatory and collision-free robot navigation in densely populated human environments.

VI. Future Work

While the Stochastic GAT successfully models interaction uncertainty via Monte Carlo sampling, it relies on manually injected perturbations. In future work, we plan to replace this surrogate stochasticity with a fully generative **Flow Matching** architecture. Flow Matching will allow us to directly learn the continuous probability density vector fields of pedestrian trajectories, providing mathematically rigorous multi-modal generation without modifying the underlying coordinate history.

Additionally, we aim to extend our experimental validation into the control domain. We will actively integrate the proposed SLAM pipeline with an advanced Model Predictive Controller (MPC) within the simulator to empirically measure collision rates and navigation efficiency. This will conclusively validate how the extracted covariance ellipses enhance the robot's real-time safety in reactive, multi-agent scenarios.

Figure 4: Demonstrates qualitative superiority of stochastic GAT over deterministic GAT - the stochastic version with Mahalanobis loss produces smooth predictions aligned with ground truth, while deterministic version appears jittery and divergent.

Limitations

- Evaluated only in 2D simulation with pyminisim, not validated on real-world robotic systems
- Relies on manually injected Gaussian perturbations rather than learned stochastic generation
- No integration with actual MPC/PPO controllers to validate claimed safety benefits
- Limited to pedestrian dynamics, unclear how approach generalizes to other dynamic entities like vehicles